

Exercise 1: Calculate Fourier series of the function f defined on $\mathbb{R}$ by

$$f(x) = \cos^3(x).$$

Exercise 2: Calculate Fourier series of the function f that equals 1 on $]0, \pi[$ and -1 on $] - \pi, 0[$. Then deduce the following sums:

$$\sum_{n=0}^{\infty} (-1)^n \frac{1}{2n+1}, \quad \sum_{n=0}^{\infty} \frac{1}{(2n+1)^2}$$

Exercise 3: Calculate Fourier series of the function f such that, on $[-\pi, \pi]$,

$$f(x) = x(\pi - x)(\pi + x).$$

Then deduce the following sums:

$$\sum_{n=0}^{\infty} (-1)^n \frac{1}{(2n+1)^3}, \quad \sum_{n=1}^{\infty} \frac{1}{n^6}$$

Exercise 4: Let a be a number such that $0 < a < \pi$. Define two elements f_a and g_a in E as follows:

$$f_a(x) = \begin{cases} 0 & \text{on } [0, \pi - a] \\ 1 & \text{on }]\pi - a, \pi + a[\\ 0 & \text{on } [\pi + a, 2\pi] \end{cases}, \quad g_a(x) = \begin{cases} 0 & \text{on } [-\pi, -a[\\ 1 & \text{on } [-a, a] \\ 0 & \text{on }]a, \pi] \end{cases}$$

Check that $g_a = 1 - f_{\pi-a}$, and calculate Fourier series of f_a and g_a . Then deduce the following sums:

$$\sum_{n=1}^{\infty} \frac{\sin^2(na)}{n^2}, \quad \sum_{n=0}^{\infty} \frac{1}{(2n+1)^2}, \quad \sum_{n=1}^{\infty} \frac{\sin(na)}{n}, \quad \sum_{n=1}^{\infty} \frac{(-1)^n \sin(na)}{n}$$

Exercise 5: Consider the function $f : [-1; 1] \rightarrow \mathbb{R}$, a 2-periodic function defined as:

$$f(x) = \begin{cases} x - x^3 & \text{if } x \in [-1; 1] \end{cases}$$

1. Plot the graph of f in the interval $[-2; 2]$.
2. Calculate the Fourier coefficients of f and then provide its Fourier series.
3. Deduce the sum of the following numerical series:

$$S_1 = \sum_{n=0}^{\infty} (-1)^n \frac{1}{(2n+1)^3}, \quad S_2 = \sum_{n=1}^{\infty} \frac{1}{n^6}$$